SC01

Intelligent Campus Climate & Energy Controller

Soft Computing Capstone — Final Presentation

Vansh Sharma • Manshi Tiwari • Lakshya Sharma • Akash Dubey

Simulation-based evaluation • Synthetic dataset • Streamlit interface

1. Problem & Objectives

What the system is designed to solve

- **Problem:** Campus facility managers need understandable room-climate recommendations that account for temperature, humidity, occupancy and electricity tariff.
- **Goal:** Recommend cooling intensity, fan level and an energy-management action.
- **Methods:** deterministic baseline → fuzzy inference → GA-tuned fuzzy controller.
- **Evidence:** reproducible synthetic scenarios, validation, comparative evaluation, visualizations and an interactive Streamlit interface.

Important scope: this is a software simulation. The outputs are recommendations, not measured electricity consumption or verified HVAC savings.

2. System Inputs, Outputs & Workflow

Input	Meaning	Domain
temperature_c	Room temperature	18–45 °C
humidity_pct	Relative humidity	20–100%
occupancy_count	Number of occupants	0–100
tariff_level	Electricity tariff	low / medium / high

Output	Meaning	Values
cooling_pct	Recommended cooling intensity	0–100%
fan_level	Recommended fan level	off / low / medium / high
energy_action	Energy-management recommendation	normal / save / avoid_peak

Workflow: Scenario generation → validation → baseline controller → fuzzy inference → GA tuning → common-scenario comparison → visualization → Streamlit interface

3. Dataset & Reproducibility

- **10,000** synthetic climate/occupancy scenarios.
- Random seed: **42** for repeatability.
- Five input columns with separate raw and processed datasets.
- Recorded checks: **0 missing values**, **0 duplicate scenario IDs**, **0 duplicate complete rows**.
- Recorded ranges: temperature 18.00–44.99 °C; humidity 20.01–99.99%; occupancy 0–100.
- Tariff distribution: 3,519 low • 3,506 medium • 2,975 high.

Why synthetic? The project is intended as a reproducible simulation. Real-campus sensor data and measured HVAC electricity use were not available for this evaluation.

4. Controller Methods

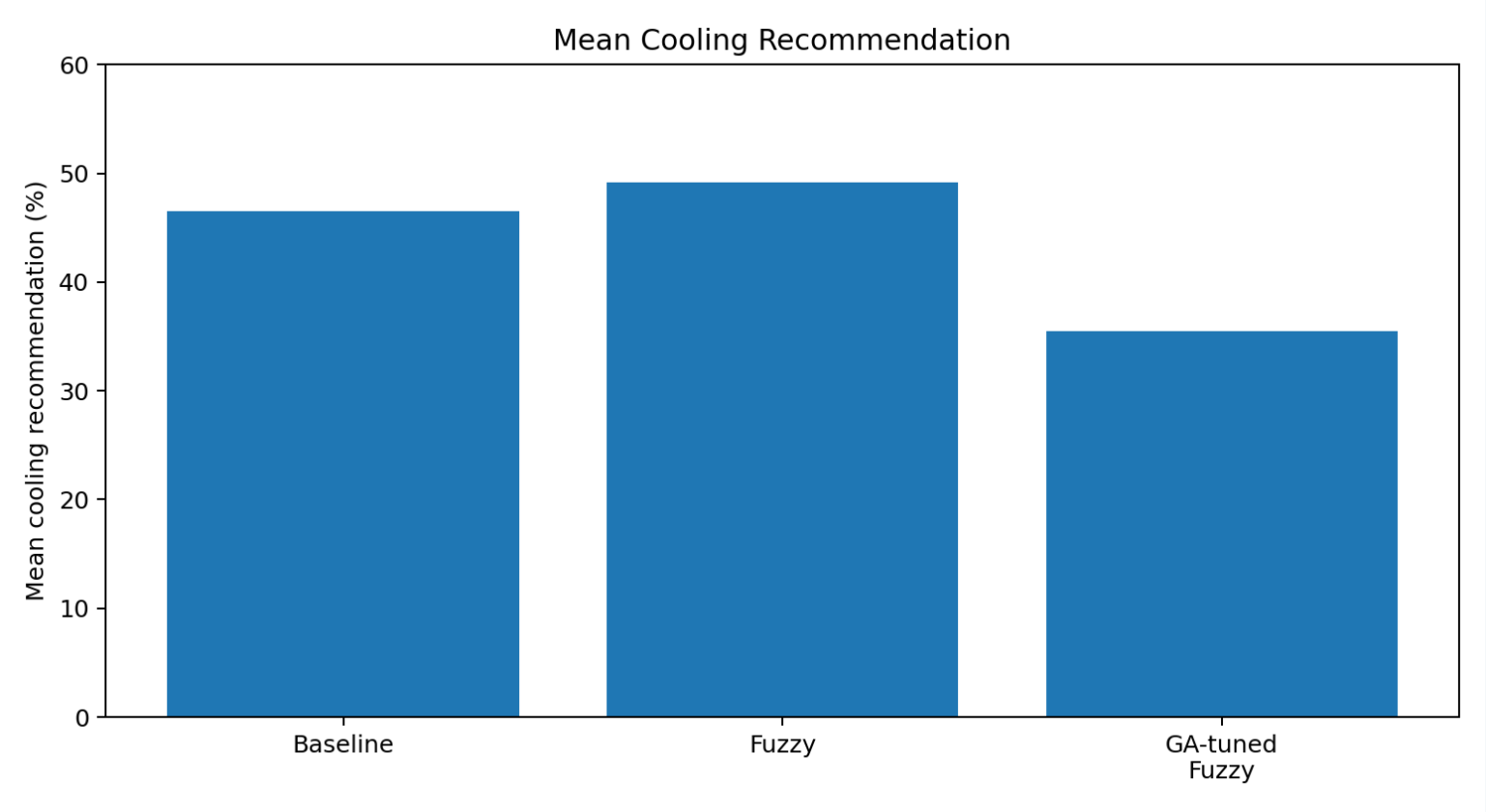
Method	Core idea	Role in evaluation
Rule-based baseline	Fixed temperature/occupancy/tariff rules	Reference controller
Fuzzy controller	Membership functions + fuzzy inference	Gradual response to conditions
GA-tuned fuzzy	Genetic algorithm searches controller parameters	Parameter optimization against a synthetic objective

GA configuration: population 24 • 20 generations • 400 training samples • seeded evolutionary process. The fitness target is a synthetic cooling-demand heuristic with a small high-tariff penalty.

Interpretation: a lower GA fitness value is evidence of a better fit to that synthetic objective—not proof of comfort, safety or real energy efficiency.

5. Results — Mean Cooling Recommendation

Recorded three-controller evaluation: 20 scenarios

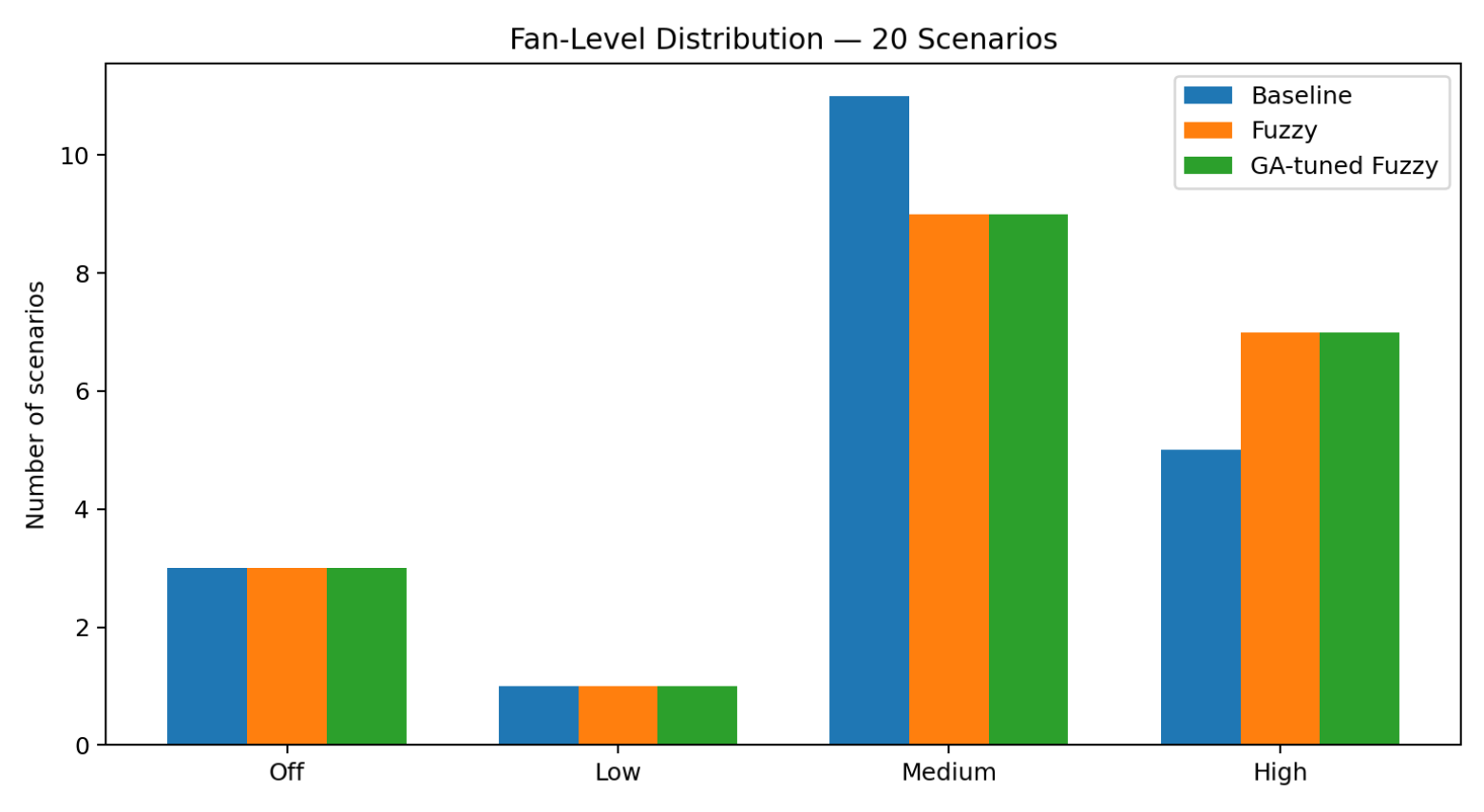


Controller	Mean cooling
Baseline	46.50%
Fuzzy	49.15%
GA-tuned fuzzy	35.46%

The GA-tuned controller produced the lowest mean cooling recommendation in the recorded 20-scenario evaluation. This is a simulated control output, not a measured energy result.

6. Results — Fan-Level Distribution

Counts across the same 20 evaluation scenarios

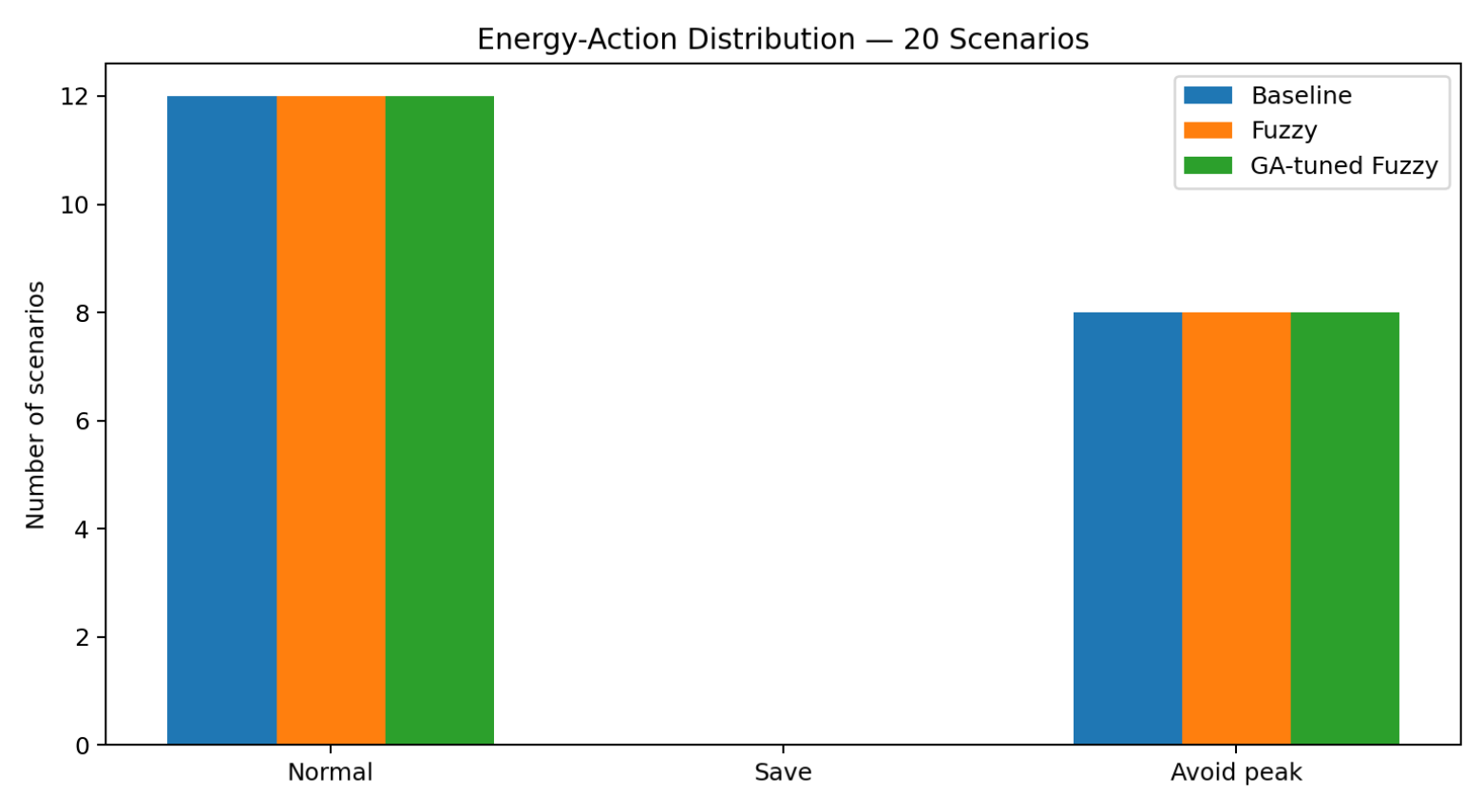


Fan level	Baseline	Fuzzy	GA
Off	3	3	3
Low	1	1	1
Medium	11	9	9
High	5	7	7

The fuzzy-based methods shift some scenarios from medium toward high fan output, while preserving the same off/low counts in this evaluation set.

7. Results — Energy-Action Distribution

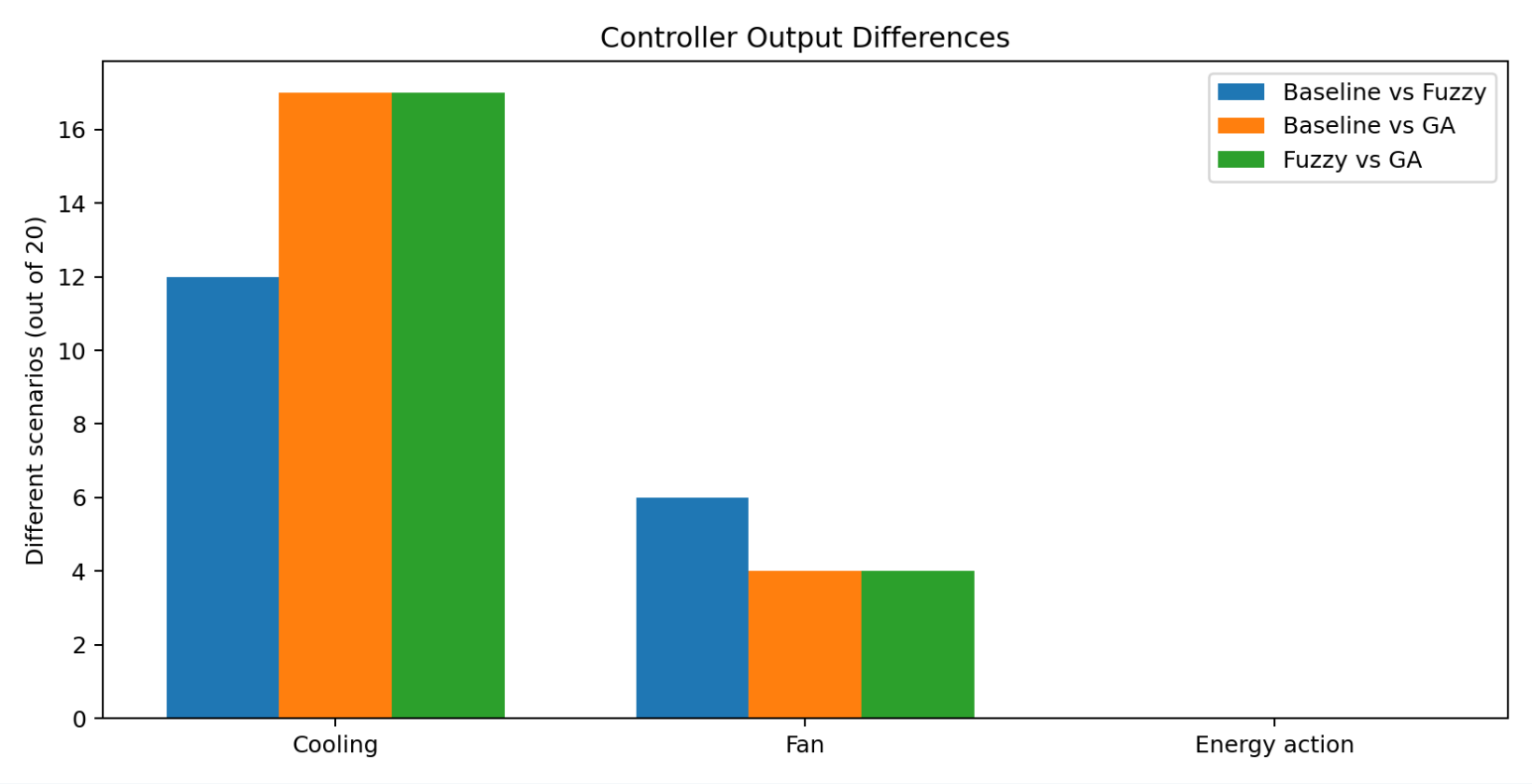
All three controllers agreed on these recorded 20 scenarios



Action	Baseline	Fuzzy	GA
Normal	12	12	12
Save	0	0	0
Avoid peak	8	8	8

Energy-action outputs were identical across the three controllers for the 20 recorded scenarios.

8. Results — Where the Controllers Differ



Comparison	Cooling	Fan
Baseline vs Fuzzy	12/20	6/20
Baseline vs GA	17/20	4/20
Fuzzy vs GA	17/20	4/20

These differences show that the controllers behave differently under the same scenarios. They do not establish that one controller is operationally superior.

9. Interactive Product V1 — Streamlit Dashboard

User flow

- Enter temperature, humidity, occupancy and tariff.
- Run the controller recommendations.
- Review cooling, fan and energy-action outputs.
- Use the dashboard as the demonstration interface.

Streamlit interface

Temperature Humidity Occupancy Tariff → Cooling recommendation → Fan level → Energy action

Deployment: Streamlit Community Cloud (browser-tested previously; recheck the public URL immediately before submission).

10. Validation, Limitations & Evidence

- **Validation:** input fixtures, scenario coverage, fuzzy-controller boundary tests, full-dataset evaluation and paired controller comparison.
- **Fair comparison:** controllers are evaluated on common scenarios; baseline and fuzzy comparison is paired by scenario ID.
- **Reproducibility:** fixed dataset seed and recorded result CSVs.
- **Limitation:** synthetic scenarios and heuristic cooling targets; no measured kWh, thermal-comfort score or live HVAC telemetry.
- **GA limitation:** optimization target is synthetic and should not be interpreted as a validated physical energy model.
- **Submission evidence:** code, CSV results, documentation, dashboard and technical report are available in the repository.

11. Conclusion & Viva Takeaways

- **What was built:** reproducible data pipeline + deterministic baseline + fuzzy controller + GA-tuned fuzzy controller + comparative evaluation + dashboard.
- **What the experiments show:** controller outputs differ meaningfully while all methods respond to the same scenario inputs.
- **Strongest quantitative point:** GA-tuned fuzzy mean cooling = **35.46%** vs baseline **46.50%** on the recorded 20-scenario evaluation.
- **Do not overclaim:** lower simulated cooling is not equivalent to measured energy savings.
- **Future work:** measured campus data, validated comfort/energy metrics, broader stress testing, and stronger deployment/user-testing evidence.

Viva line: “Our results demonstrate controller behavior within a reproducible simulation; they are not a claim of real-world HVAC energy savings.”